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CST476 MOBILE COMPUTING

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Wireless LAN

Wireless LAN stands for Wireless Local Area Network. It is also called LAWN (Local Area Wireless Network).

WLAN is one in which a mobile user can connect to a Local Area Network (LAN) through a wireless connection.

Wireless LANs (WLANs) are wireless computer networks that use high-frequency radio waves instead of cables for connecting the devices within a limited area forming LAN (Local Area Network). Users connected by wireless LANs can move around within this limited area such as home, school, campus, office building, railway platform, etc.

Most WLANs are based upon the standard IEEE 802.11 standard or WiFi.

Types of WLANS

WLANs, as standardized by IEEE 802.11, operates in two basic modes, infrastructure, and ad hoc mode.

☐ **Infrastructure Mode** - Mobile devices or clients connect to an access point (AP) that in turn connects via a bridge to the LAN or Internet. The client transmits frames to other clients via the AP.

☐ **Ad Hoc Mode** - Clients transmit frames directly to each other in a peer-to-peer fashion.

Advantages of WLANs

- Flexibility: Within radio coverage, nodes can communicate without further restriction. Radio waves can penetrate walls, senders and receivers can be placed anywhere (also non-visible, e.g., within devices, in walls etc.).
- Planning: Only wireless ad-hoc networks allow for communication without previous planning, any wired network needs wiring plans.
- Design: Wireless networks allow for the design of independent, small devices which can for example be put into a pocket. Cables not only restrict users but also designers of small notepads, PDAs, etc.
- Robustness: Wireless networks can handle disasters, e.g., earthquakes, flood etc. whereas, networks requiring a wired infrastructure will usually break down completely in disasters.
- Cost: The cost of installing and maintaining a wireless LAN is on average lower than the cost of installing and maintaining a traditional wired LAN, for two reasons. First, after providing wireless access to the wireless network via an access point for the first user, adding additional users to a network will not increase the cost. And second, wireless LAN eliminates the direct costs of cabling and the labor associated with installing and repairing it.
- Ease of Use: Wireless LAN is easy to use and the users need very little new information to take advantage of

WLANs.

- The system is portable within the network coverage. Access to the network is not bounded by the length of the cables.
- Installation and setup are much easier than wired counterparts.
- The LANs are scalable in nature, i.e. devices may be added or removed from the network at greater ease than wired LANs.

Disadvantages of WLANs □ **Quality of Services:** Quality of wireless LAN is typically lower than wired networks.

The main reason for this is the lower bandwidth due to limitations in radio transmission, higher error rates due to interference and higher delay/delay variation due to extensive error correction and detection mechanisms.

□ **Proprietary Solutions:** Due to slow standardization procedures, many companies have come up with proprietary solutions offering standardization functionality plus many enhanced features. Most components today adhere to the basic standards IEEE 802.11a or 802.11b.

□ **Restrictions:** Several govt. and non-govt. institutions world-wide regulate the operation and restrict frequencies

to minimize interference.

- Global operation: Wireless LAN products are sold in all countries so, national and international frequency regulations have to be considered.
- Low Power: Devices communicating via a wireless LAN are typically power consuming, also wireless devices running on battery power. Whereas the LAN design should take this into account and implement special power saving modes and power management functions.
- License free operation: LAN operators don't want to apply for a special license to be able to use the product. The equipment must operate in a license free band, such as the 2.4 GHz ISM band.
- Robust transmission technology: If wireless LAN uses radio transmission, many other electrical devices can interfere with them (such as vacuum cleaner, train engines, hair dryers, etc.). Wireless LAN transceivers cannot be adjusted for perfect transmission in a standard office or production environment.
- Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.
- WLANs are slower than wired LANs.

Wireless LAN-

Mobile Terminals (MTs), on the other hand, are more powerful, and can be accessed when they are in motion.

WLANs aim to support truly mobile work stations.

WLAN Uses

Wireless computer networks are capable of offering versatile functionalities. WLANs are very flexible and can be configured in a variety of typologies based on the application. Some possible uses of WLANs are described below.

- ☐ Users would be able to surf the Internet, check e-mail, and receive Instant Messages on the move.
- ☐ In areas affected by earthquakes or other disasters, no suitable infrastructure may be available on the site.

WLANs are handy in such locations to set up networks on the fly.

- ☐ There are many historic buildings where there has been a need to set up computer networks. In such places, wiring may not be permitted or the building design may not be conducive to efficient wiring. WLANs are very good solutions in such places.

Design Goals

The following are some of the goals which have to be achieved while designing WLANs:

- **Operational simplicity** - Design of wireless LANs must incorporate features to enable a mobile user to quickly set up and access network services in a simple and efficient manner.
- **Power efficient operation** - The power-constrained nature of mobile computing devices such as laptops and PDAs necessitates the important requirement of WLANs operating with minimal power consumption. Therefore, the design of WLAN must incorporate power-saving features and use appropriate technologies and protocols to achieve this.
- **License-free operation** - One of the major factors that affects the cost of wireless access is the license fee for the spectrum in which a particular wireless access technology operates. Low cost of access is an important aspect for popularizing a WLAN technology. Hence the design of WLAN should consider the parts of the frequency spectrum. For its operation which does not require an explicit.
- **Tolerance to interference** - The proliferation of different wireless networking technologies both for civilian and

military applications have led to a significant increase in the interference level across the radio spectrum.

□ **The WLAN design** should account for this and take appropriate measures by way of selecting technologies and protocols to operate in the presence of interference.

□ **Global Usability** - The design of the WLAN, the choice of technology, and the selection of the operating frequency spectrum should take into account the prevailing spectrum restriction in countries across the world. This ensures the acceptability of the technology across the world.

□ **Security** - The inherent broadcast nature of wireless medium adds to the requirement of security features to be included in the design of WLAN technology.

□ **Safety requirements** - The design of WLAN technology should follow the safety requirements that can be classified into the following.

□ Interference to medical and other instrumentation devices.

□ Increased power level of transmitters that can lead to health hazards.

Network Architecture

Network architecture describes the types of WLANs, the components of a typical WLAN and the services offered by a WLAN. Infrastructure based versus Ad Hoc LANs.

WLANs can be broadly classified into two types, namely Infrastructure networks and Ad hoc LANs, based on the underlying architecture.

Infrastructure networks

Infrastructure networks contain special nodes called Access Points (APs), which are connected via existing networks.

☐ APs are special in the sense that they can interact with wireless nodes as well as with the existing wired network.

☐ The other wireless nodes, also known as Mobile stations (STAs), communicate via APs.

☐ The APs also act as bridges with other networks.

Ad-hoc LANs

Ad hoc LANs do not need any fixed infrastructure. These networks can be set up on the fly at any place. Nodes communicate directly with each other for forward messages through other nodes that are directly accessible.

Wireless LAN and IEEE 802.11

Wireless LANs are those Local Area Networks that use high frequency radio waves instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage. Most WLANs are based upon the standard IEEE 802.11 or WiFi.

IEEE 802.11 Architecture

The components of an IEEE 802.11 architecture are as follows

1) Stations (STA) - Stations comprise all devices and equipments that are connected to the wireless LAN. A station can be of two types:

- ☐ Wireless Access Pointz (WAP) - WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.

- ☐ Client - Clients are workstations, computers, laptops, printers, smartphones, etc.

Each station has a wireless network interface controller.

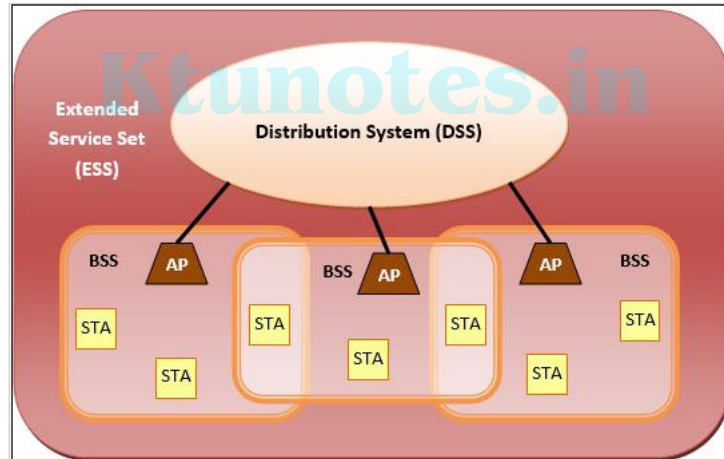
2) Basic Service Set (BSS) - A basic service set is a group of stations communicating at physical layer level.

BSS can be of two categories depending upon mode of operation:

- ☐ Infrastructure BSS - Here, the devices communicate with other devices through access points.
- ☐ Independent BSS ; Here, the devices communicate in peer-to-peer basis in an ad hoc manner

3) Extended Service Set (ESS) - It is a set of all connected BSS.

4) Distribution System (DS) - It connects access points in ESS.



Advantages of WLANs

1. They provide clutter free homes, offices and other networked places.
2. The LANs are scalable in nature, i.e. devices may be added or removed from the network at a greater ease than wired LANs.
3. The system is portable within the network coverage and access to the network is not bounded by the length of the cables.
4. Installation and setup is much easier than wired counterparts.
5. The equipment and setup costs are reduced.

Disadvantages of WLANs

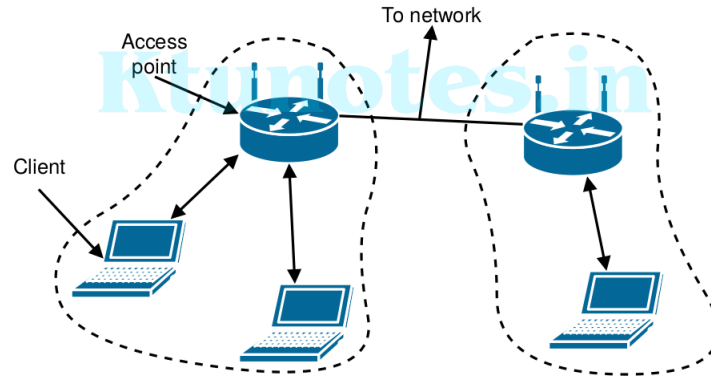
1. Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.
2. Greater care is needed for encrypting information. Also, they are more prone to errors. So, they require greater bandwidth than the wired LANs.
3. WLANs are slower than wired LANs.

802.11

802.11, commonly known as WiFi, is a family of wireless protocols standardized by IEEE.

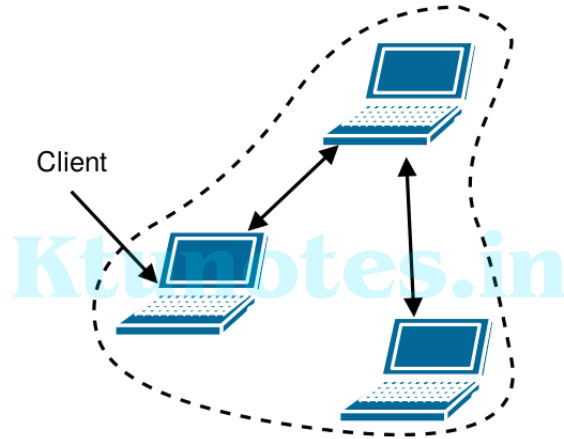
802.11 networks can be used in two modes: infrastructure mode and ad-hoc mode.

Infrastructure mode uses an AP (Access Point) that is connected to the network. Clients send and receive packets via the AP. Several APs can be connected to form an extended network.



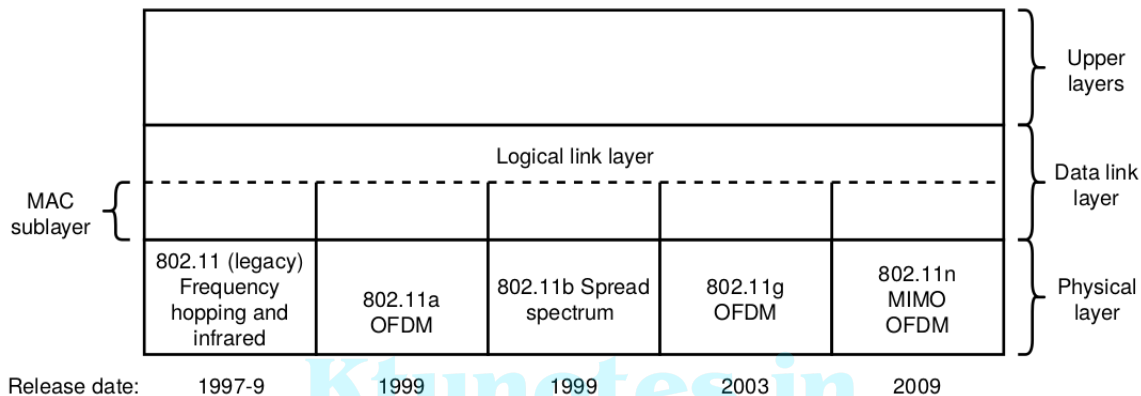
802.11 architecture in infrastructure mode

Ad-hoc mode is a collection of computers connected to each other so that they can send frames to each other. There's no AP. The 802.11 physical layer corresponds to the OSI physical layer, but the data link layer is split into



802.11 architecture in ad-hoc mode

multiple sublayers. In 802.11 the MAC sublayer determines which channel gets to transmit next. The sublayer above, the LLC (Logical Link Layer), hides the differences between the varying 802.11 versions for the network layer



802.11 protocol stack

The 802.11 physical layer

All 802.11 techniques use short-range radios to transmit signals in either 2.4-GHz or 5-GHz ISM frequency bands. These bands are unlicensed, and so are shared by many other devices such as garage door openers, or microwave ovens. Fewer applications tend to use the 5-GHz band, so 5-GHz can be better for some applications despite shorter range due to higher frequency.

All 802.11 transmission methods define multiple rates. Different rates can be used depending on the current conditions. If the signal is weak, a low rate is used. If the signal is clear, the highest rate is used. The process of adjustment is called rate adaption.

802.11b

802.11b is a spread-spectrum method. It supports rates of 1, 2, 5.5, and 11Mb/s. 802.11b is similar to the CDMA system, except that one spreading code is shared between all users. 802.11b uses a spreading sequence called the Barker Sequence. The auto correlation of the Barker Sequence is low except when sequences are aligned. This allows a receiver to lock onto the start of a transmission.

802.11a

802.11a was standardized after 802.11b, despite the group being formed first. It supports rates up to 54Mb/s in the 5-GHz ISM band. 802.11a is based on OFDM (Orthogonal Frequency Division Multiplexing). Bits are sent over 52 sub-carriers in parallel. 48 carry data, and 4 are used for synchronization. A symbol lasts $4\frac{1}{4}$ μs, and sends either 1, 2, 4, or 6 bits. The bits are coded for error correction with a binary convolutional code first so only 1/2, 2/3, or 3/4 of the bits are not redundant. 802.11a can run at different rates using the different combinations . The rates range

from 5 to 55Mb/s.

802.11g

802.11g uses the OFDM modulation methods of 802.11a, but operates in 2.4GHz ISM band. It has the same rates as 802.11a, as well as compatibility with 802.11b devices.

802.11n

802.11n was ratified in 2009. The aim of 802.11n was throughput of 100Mb/s after transmission overheads were removed.

Highlights:

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1. Channels were doubled from 20MHz to 40MHz.
2. Frame overhead was reduced by allowing a group of frames to be sent together.
3. Up to four streams could be transmitted at a time using four antennas.

In 802.11, the stream signals interfere at the receiver, but they can be separated using MIMO (Multiple Input Multiple Output) techniques.

The MAC sublayer protocol

The 802.11 MAC sublayer is different from the Ethernet MAC sublayer for two reasons:

- Radios are almost always half duplex.
- Transmission ranges of different stations might be different

802.11 uses the CSMA/CA (CSMA with Collision Avoidance) protocol. CSMA/CA is similar to ethernet CSMA/CD. It uses channel sensing and exponential backoff after collisions, but instead of entering backoff once a collision has been detected, CSMA/CA uses backoff immediately.

The algorithm will backoff for a number of slots, for example 0 to 15 in the case of the OFDM physical layer. The station waits until the channel is idle by sensing that there is no signal for a short period of time. It counts down idle slots, pausing when frames are sent. When its counter reaches 0, it sends its frames. Acknowledgements are used to infer collisions because collisions cannot be detected.

This way of operating is called DCF (Distributed Coordination Function), in DCF each station is acting independently, without a central control. The other problem facing 802.11 protocols is transmission ranges differing

between stations. It's possible for transmissions in one part of a cell to not be received in another part of the cell, which can make it impossible for a sender to sense a busy channel, resulting in collisions. 802.11 defines channel sensing to consist of physical and virtual sensing. Physical sensing checks the medium to see if there is a valid signal.

With virtual sensing, each station keeps a record of what channel is in use. It does this with the NAV (Network Allocation Vector). Each frame includes a NAV field that contains information on how long the sequence that the frame is part of will take to complete.

The main strategy for reliability is to lower the transmission rate if too many frames are unsuccessful. Lower transmission rates use more robust modulations. If too many frames are lost, a station can lower its rate. If frames are successfully delivered, a station can test a higher rate to see if it should upgrade.

Another strategy for successful transmissions is to send shorter frames. 802.11 allows frames to be split into fragments, with their own checksum. The fragment size can be adjusted by the AP. Fragments are numbered and sent using a stop-and-wait protocol.

802.11 uses beacon frames. Beacon frames are broadcast periodically by the AP. The frames advertise the presence of the AP to clients and carry system parameters, such as the identifier of the AP, the time, how long until the next beacon, and security settings.

Clients can set a power-management bit in frames that are sent to the AP to alert it that the client is entering power-save mode. In power-save mode, the client rests and the AP buffers traffic intended for it. The client wakes up for every beacon, and checks a traffic map that's sent with the beacon. The traffic map tells the client whether there is buffered traffic. If there is, the client sends a poll to the AP, and the AP sends the buffered traffic.

802.11 provides quality of service by extending CSMA/CA with defined intervals between frames. Different kinds of frames have different time intervals. The interval between regular data frame is called the DIFS (DCF InterFrame Spacing). Any station can attempt to acquire a channel after the channel has been idle for DIFS.

The shortest interval is SIFS (Short InterFrame Spacing). SIFS is used to send an ACK, other control frames like RTS, or for sending another fragment.

Different priorities of traffic are determined with different AIFS (Arbitration InterFrame Space) intervals. A short AIF can allow the AP to send higher priority traffic. An AIF that is longer than DIFS means the traffic will be sent after regular traffic.

Another quality of service mechanism is transmission opportunity. Previously, CSMA/CA allowed only one frame to be sent at a time. This slowed down stations with significantly faster rates. Transmission opportunities make it so each station has equal airtime, not an equal number of sent frames.

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Services

802.11 defines a number of services that must be provided by conformant wireless LANs. Mobile stations use the association service to connect to APs. Usually, the service is used just after a station has moved within range of an AP. When the station is within range, it learns the identity and capabilities of the AP through either beacon frames, or by asking the AP directly. The station sends a request to associate with the AP, which the AP can either accept or reject.

The re-association service is used to let a station change its preferred AP. If correctly used, there should be no data loss between the handover. The station or the AP can also disassociate. The station should use this before shutting down.

Stations should authenticate before sending frames via the AP. Authentication is handled differently depending on the security scheme. If the network is open, anyone can use it. Otherwise credentials are required. WPA2 (WiFi Protected Access 2) is the recommended approach that implements security defined in the 802.11i standard. With WPA2, the AP communicates with an authentication server that has a username and password database to determine if the station is allowed to access the network.

The distribution service determines how to route frames from the AP. If the destination is local, the frames are sent over the air. If they are not, they are forwarded over the wired network.

The integration service handles translation for frames to be sent outside the 802.11 LAN

The data delivery service lets stations transmit and receive data using the protocols outlined in this section.

A privacy service manages encryption and decryption. The encryption algorithm for WPA2 is based on AES (Advanced Encryption Standard). The encryption keys are determined during authentication.

The QOS traffic scheduling is used to handle traffic with different priorities. It uses the protocols described in The MAC sublayer protocol section.

The transmit power control service gives stations the information they need to meet regulatory limits on transmit power that vary from region to region.

The dynamic frequency selection service give stations the information they need to avoid transmitting on frequencies in the 5-GHz band that are being used for radar in the proximity.

HIPERLAN

A high-performance local area network (HIPERLAN) is an alternative wireless LAN standard to the IEEE 802.11. It is one of four standards (HIPERLAN 1 through 4) specified by the European telecommunications standards institute (ETSI) to link interoperable technologies from different locations instead of cable. HIPERLAN uses cellular-based data networks to connect to an ATM backbone.

The main idea behind HIPERLAN is to provide an infrastructure or ad-hoc wireless system with a small radius. HIPERLAN emerged in 1991 with the goal of achieving higher data rates than the 802.11 standard. It was approved in 1996. A second version was introduced in 2000. This version is designed as a fast wireless connection

and can be used with various networks, such as UMTS backbone, ATM, and IP networks. HiperLAN/2 can also be used as a home network and supports a data rate of up to 54 Mbps.

Components of a HIPERLAN include:

- Physical Layer: This layer provides the standard functions, including radio frequency functions.
- Link Adaptation: This standard allows the access point to convey information in an uplink or downlink direction. The HIPERLAN physical layer also specifies some link adaptation algorithms to be used.
- Data Link Control (DLC) Layer: This layer includes the Media Access Control (MAC), Radio Link Control (RLC), Dynamic Frequency Selection (DFS) and Error Control (EC) protocols.
- Convergence Layer: Its basic function is to provide the HIPERLAN DLC and physical access to other data networks.

The two main differences between HiperLAN types 1 and 2 are as follows:

- Type 1 has a distributed MAC with QoS provisions, whereas type 2 has a centralized schedule MAC.

- Type 1 is based on Gaussian minimum shift keying (GMSK), whereas type 2 is based on OFDM.
- HiperLAN/2 automatically performs handoff to the nearest access point. The access point is basically a radio BS that covers an area of about 30 to 150 meters, depending on the environment. MANETs can also be created easily.

The goals of HiperLAN are as follows:

- QoS (to build multiservice network).
- Strong security.
- Handoff when moving between local area and wide areas.
- Increased throughput.
- Ease of use, deployment, and maintenance.
- Affordability.
- Scalability.

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Bluetooth

Bluetooth is a network technology that connects mobile devices wirelessly over a short range to form a personal area network (PAN). They use short-wavelength, ultra-high frequency (UHF) radio waves within the range 2.400 to 2.485 GHz, instead of RS-232 data cables of wired PANs.

There are two types of Bluetooth networks

- Piconets
- Scatternets

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Piconets

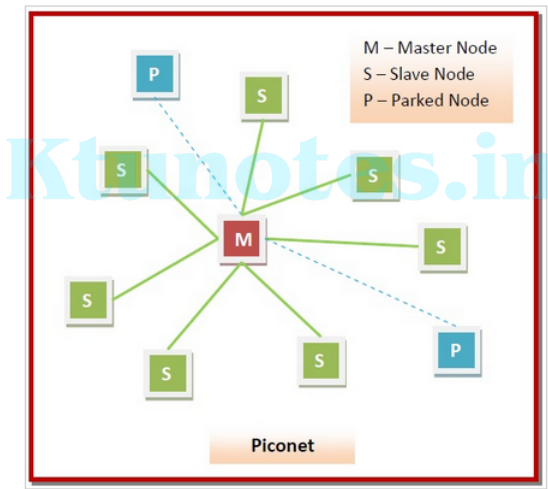
Piconets are small Bluetooth networks, formed by at most 8 stations, one of which is the master node and the rest slave nodes (maximum of 7 slaves). Master node is the primary station that manages the small network. The slave stations are secondary stations that are synchronized with the primary station.

Communication can take place between a master node and a slave node in either one-to-one or one-to-many manner. However, no direct communication takes place between slaves. Each station, whether master or slave, is

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associated with a 48-bit fixed device address.

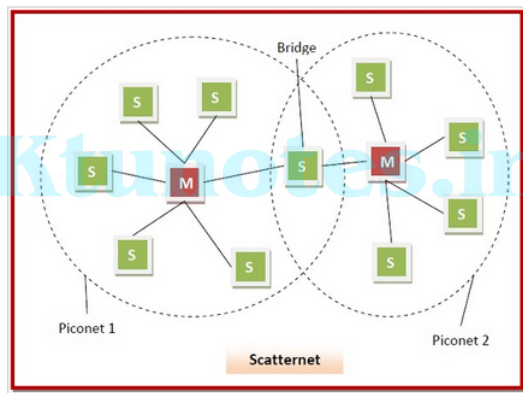
Besides the seven active slaves, there can be up to 255 numbers of parked nodes. These are in a low power state for energy conservation. The only work that they can do is respond to a beacon frame for activation from the master node.



Piconets

Scatternets

A scatternet is an interconnected collection of two or more piconets. They are formed when a node in a piconet, whether a master or a slave, acts as a slave in another piconet. This node is called the bridge between the two piconets, which connects the individual piconets to form the scatternet.



Piconets

Features of Bluetooth:

- Bluetooth Transmission capacity 720 kbps.
- Bluetooth is Wireless.
- Bluetooth is a Low-cost short-distance radio communications standard.
- Bluetooth is robust and flexible.
- Bluetooth is cable replacement technology that can be used to connect almost any device to any other device.
- The basic architecture unit of Bluetooth is a piconet.

Advantage:

- It is a low-cost and easy-to-use device.
- It can also penetrate through walls.
- It creates an Ad-hoc connection immediately without any wires.

- It is used for voice and data transfer.

Disadvantages:

- It can be hacked and hence, less secure.
- It has a slow data transfer rate: of 3 Mbps.
- It has a small range: 10 meters.
- Bluetooth communication does not support routing.
- The issues of handoffs have not been addressed.

Applications:

- It can be used in laptops, and in wireless PCs, printers.
- It can be used in wireless headsets, wireless PANs, and LANs.
- It can connect a digital camera wirelessly to a mobile phone.

- It can transfer data in terms of videos, songs, photographs, or files from one cell phone to another cell phone or computer.
- It is used in the sectors of Medical health care, sports and fitness, Military.

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